SHIP TYPES 1.2 · ABOUT 5 MIN

Deadweight, Volume and Linear-Dimension Carriers

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What Runs Out First When You Load the Ship

Deadweight carriers

Heavy cargo, SF below about 1.3 m³/t. The load line is reached with hold space left. CF below about 1.5 m³/t DWT.

Bulk and ore carriers, crude tankers, tramp cargo ships

Volume carriers

Light cargo, SF above about 2.0 m³/t. The holds are full before the load line; excess freeboard. CF above about 2.5 m³/t DWT.

Ro-Ro, car carriers, Ro-Pax, container ships, reefers, cruise ships

Linear-dimension ships

One of L, B, T or D is fixed by a canal, a port, a river or the size of the cargo unit.

Panamax ships, Great Lakes bulk carriers, cellular container ships

Special-purpose ships

Sized by a mission: bollard pull, ice class, fish hold, deck work. No general method.

Tugs, icebreakers, fishing and offshore vessels, fast craft

SF, stowage factor (the cargo): m³ needed to stow 1 t.
CF, capacity factor (the ship): hold volume / DWT, set by the designer.

Roughly: **SF below CF**, the deadweight runs out first; **SF above CF**, the holds fill first. Between 1.3 and 2.0 m³/t: either way.

When One Dimension Is Fixed from Outside

Constraint	Largest ship allowed	Fixes
Panama Canal, old locks (Panamax)	LOA 294.13 m · B 32.31 m · T 12.04 m	Beam, draught
Panama Canal, new locks (Neopanamax)	LOA 366 m · B 51.25 m · T 15.2 m	Beam, draught
St Lawrence Seaway (Seawaymax)	LOA 225.5 m · B 23.8 m · T 8.08 m	Beam, length
Azores island ports	Draught and berth length, each port	To be checked, port by port
ISO container (ISO 668)	8 ft = 2.438 m wide; 8 ft 6 in standard, 9 ft 6 in high-cube	Beam and depth, in steps

> The cost: dimensions jump in steps and the proportions suffer. A 3rd-generation Panamax container ship of about 3,700 TEU: B = 32.2 m, L = 245 m, **L/B = 7.6**.

Which Category Is the Reference Ship?

Is it the weight of the cargo or the tank volume that runs out first?

Which line of the brief fixes a main dimension from outside?

Do the ports on the route limit draught or length?

So: does the design start from deadweight or from volume?